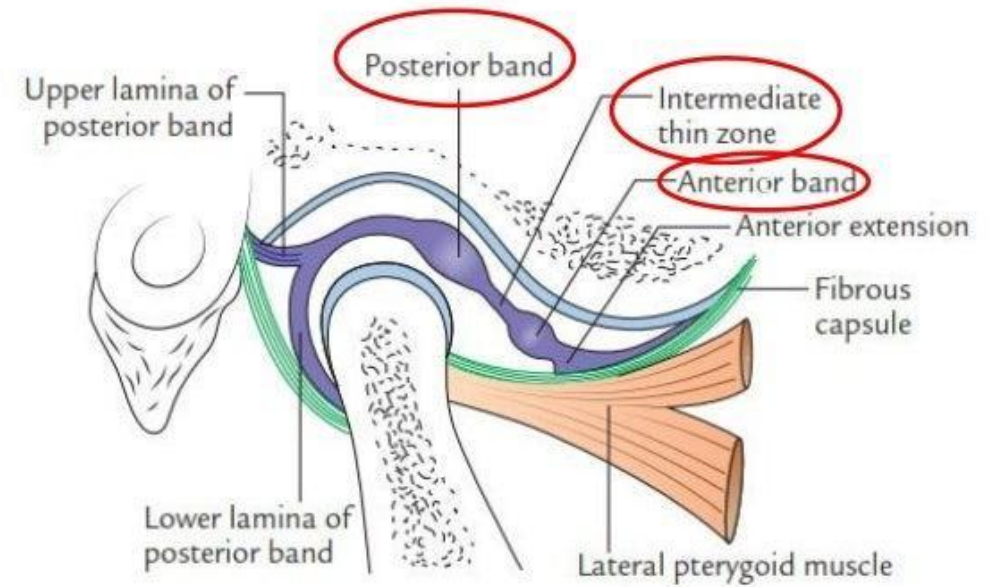


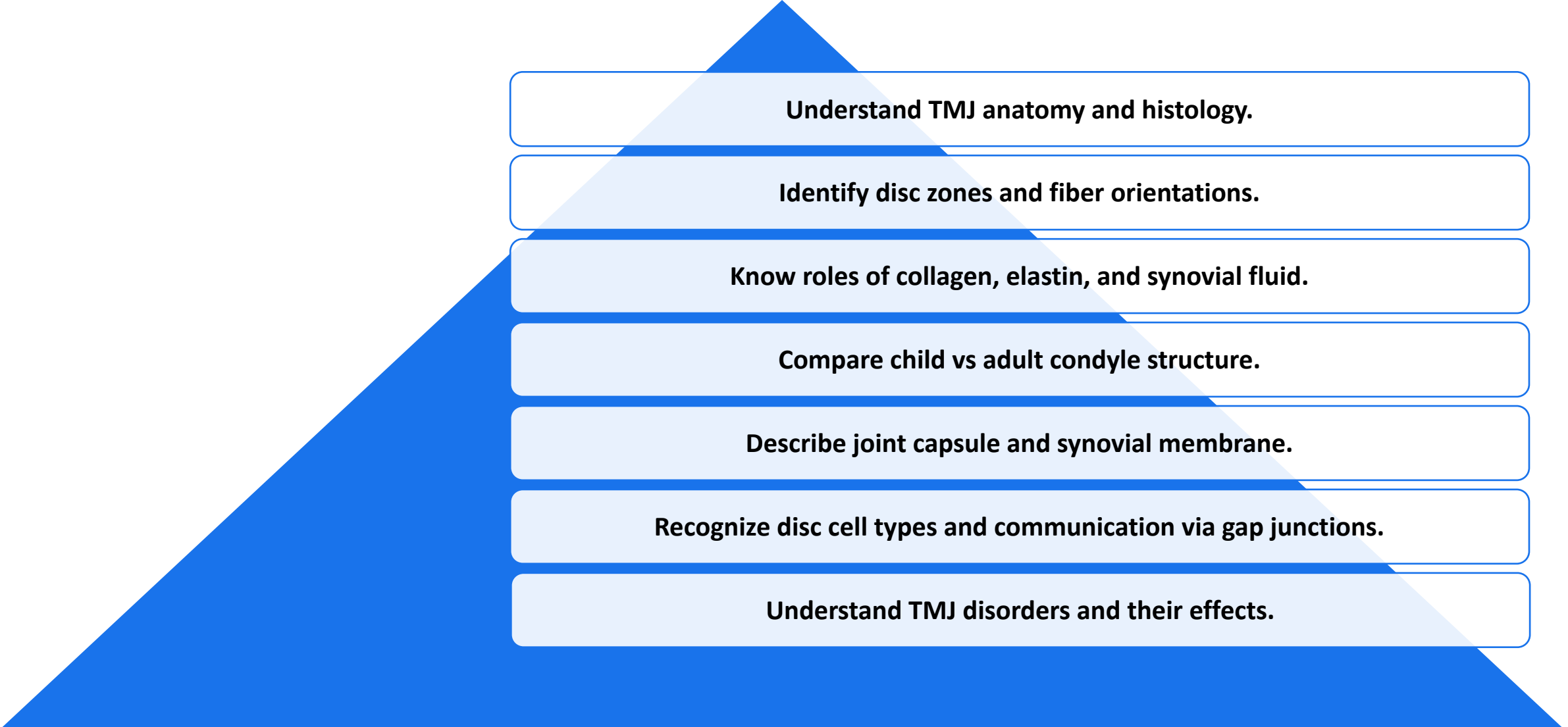
Histology of Temporomandibular Joint

Articular surfaces and disc



- DR SABA ARSHAD
- SENIOR LECTURER
- DEPARTMENT OF ORAL BIOLOGY

Learning Objectives – TMJ



Understand TMJ anatomy and histology.

Identify disc zones and fiber orientations.

Know roles of collagen, elastin, and synovial fluid.

Compare child vs adult condyle structure.

Describe joint capsule and synovial membrane.

Recognize disc cell types and communication via gap junctions.

Understand TMJ disorders and their effects.

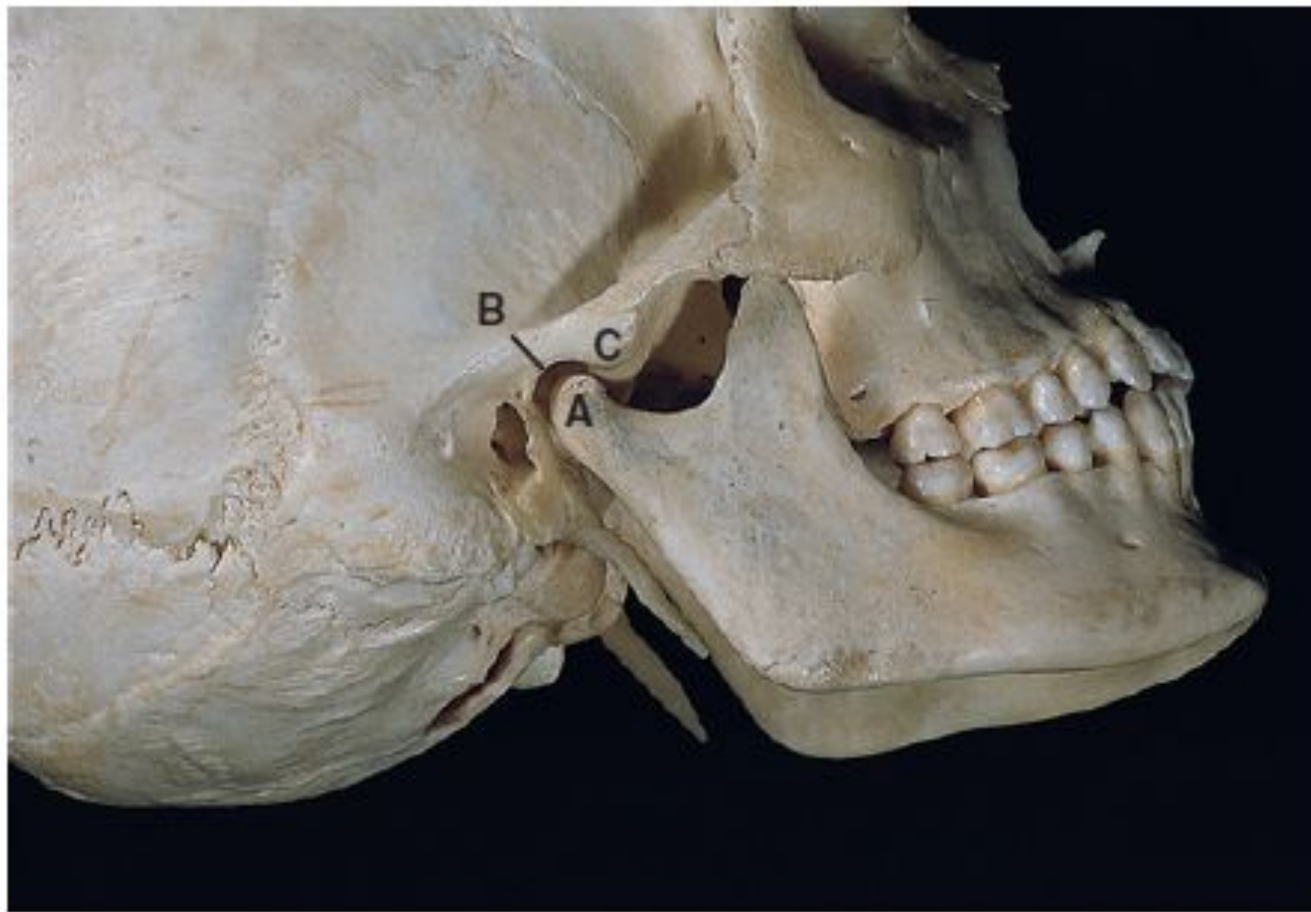
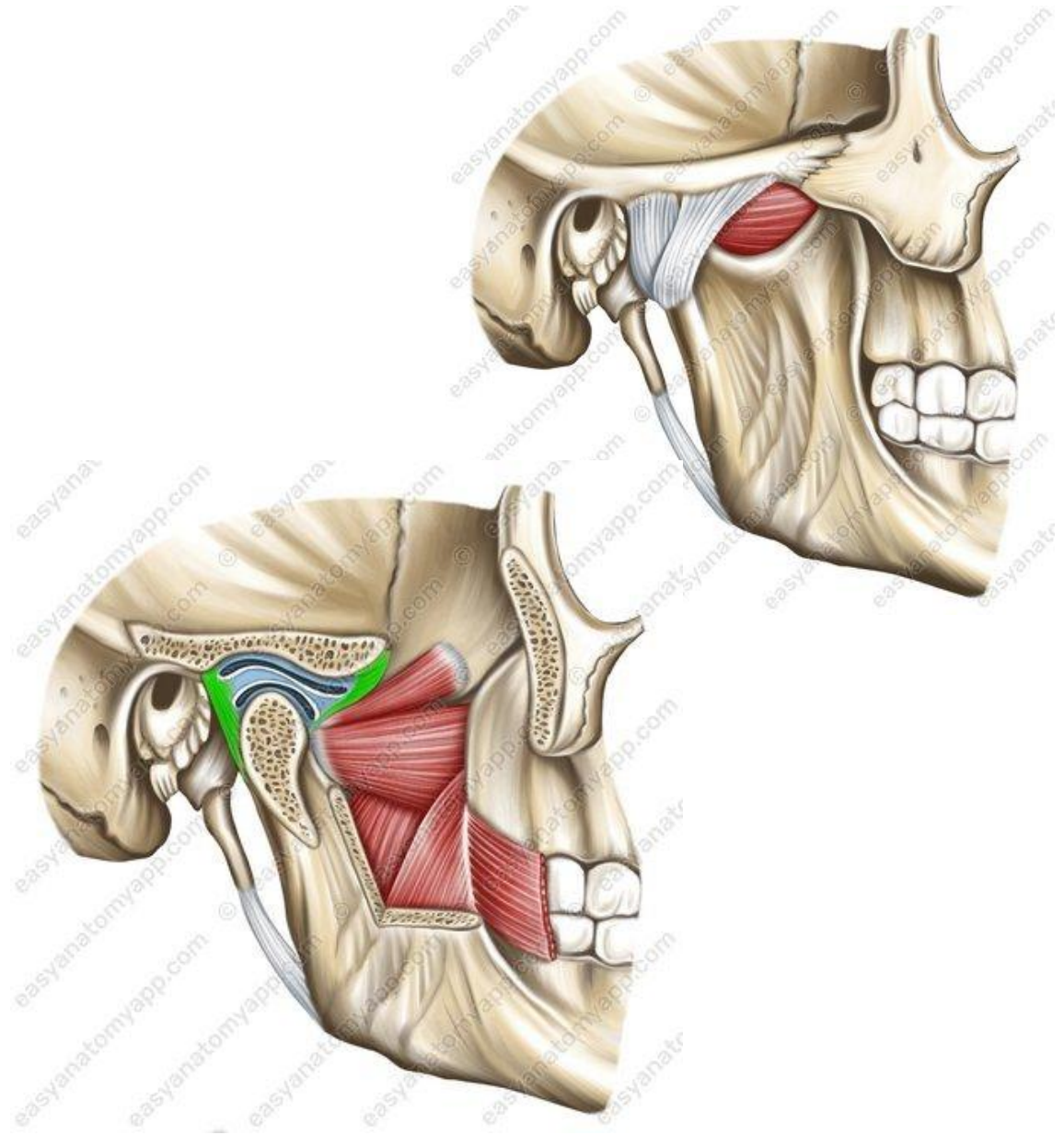


Fig. 3.1 The osteology of the temporomandibular joint. A = mandibular condyle; B = mandibular fossa of temporal bone; C = articular eminence of temporal bone.

Joint Capsule of the **TMJ**

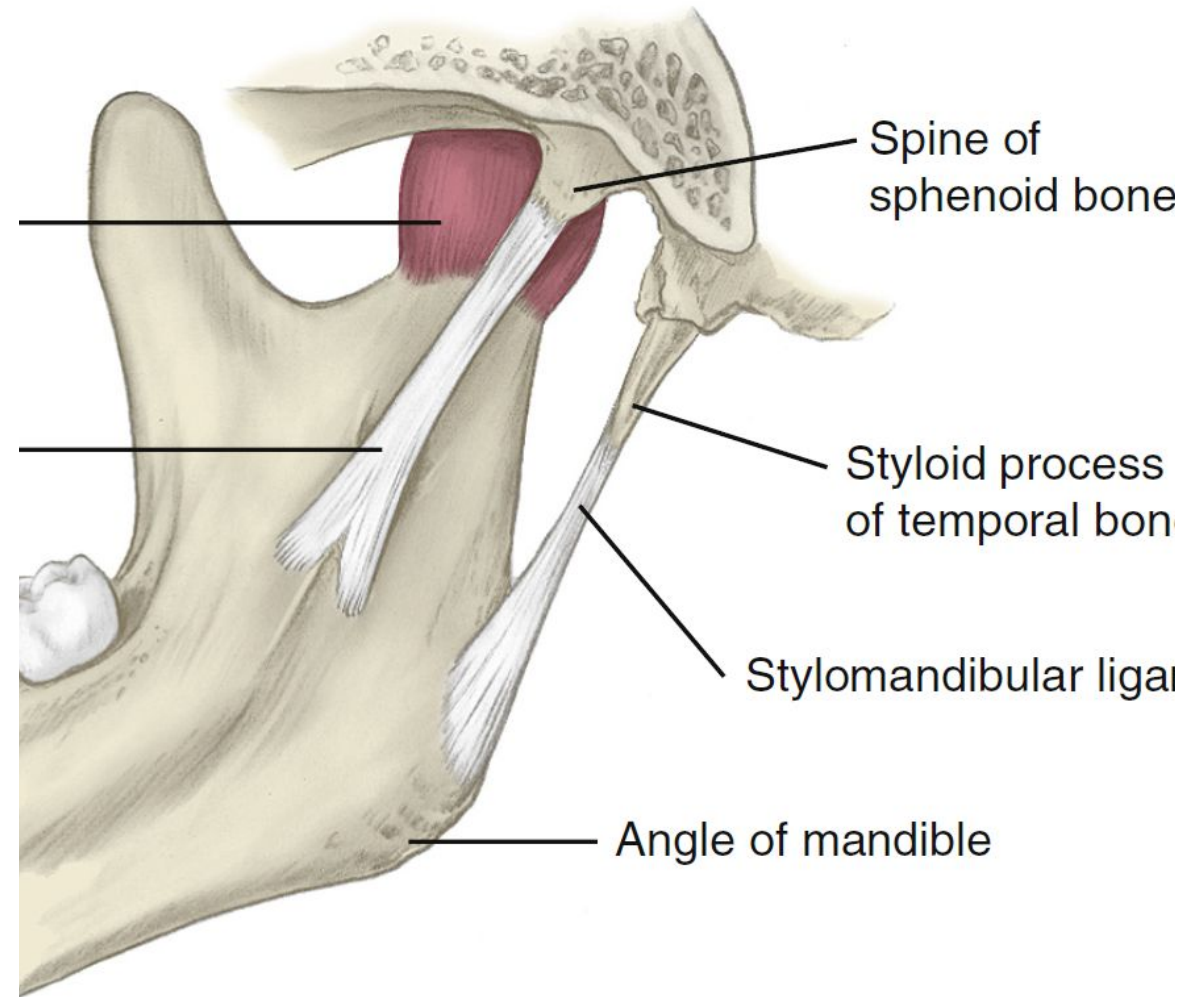
- The TMJ capsule completely encloses the joint.
- **Superior attachment:** Around the articular eminence and articular/mandibular fossa of the temporal bone.
- **Inferior attachment:** Around the neck of the mandibular condyle.

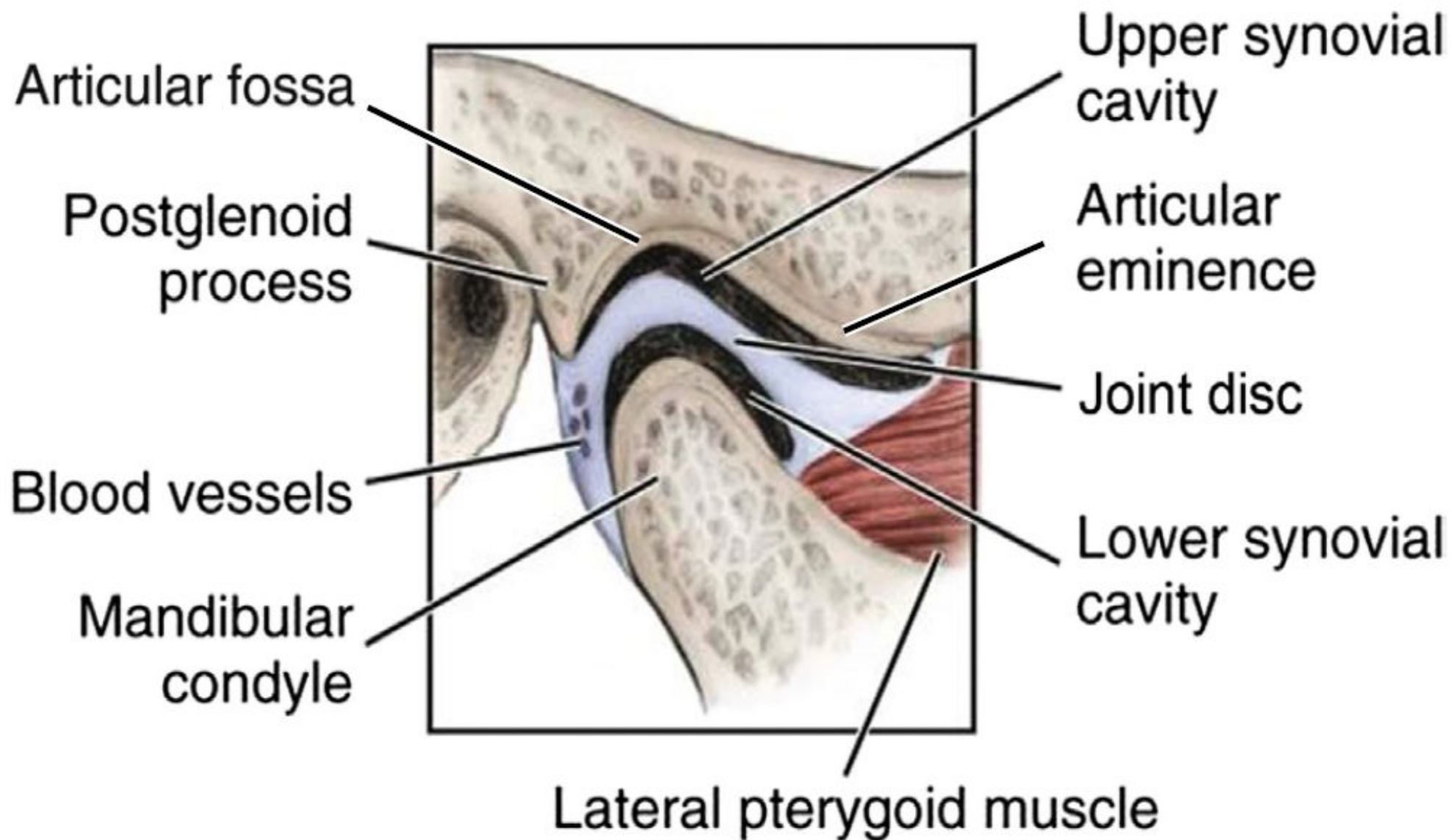


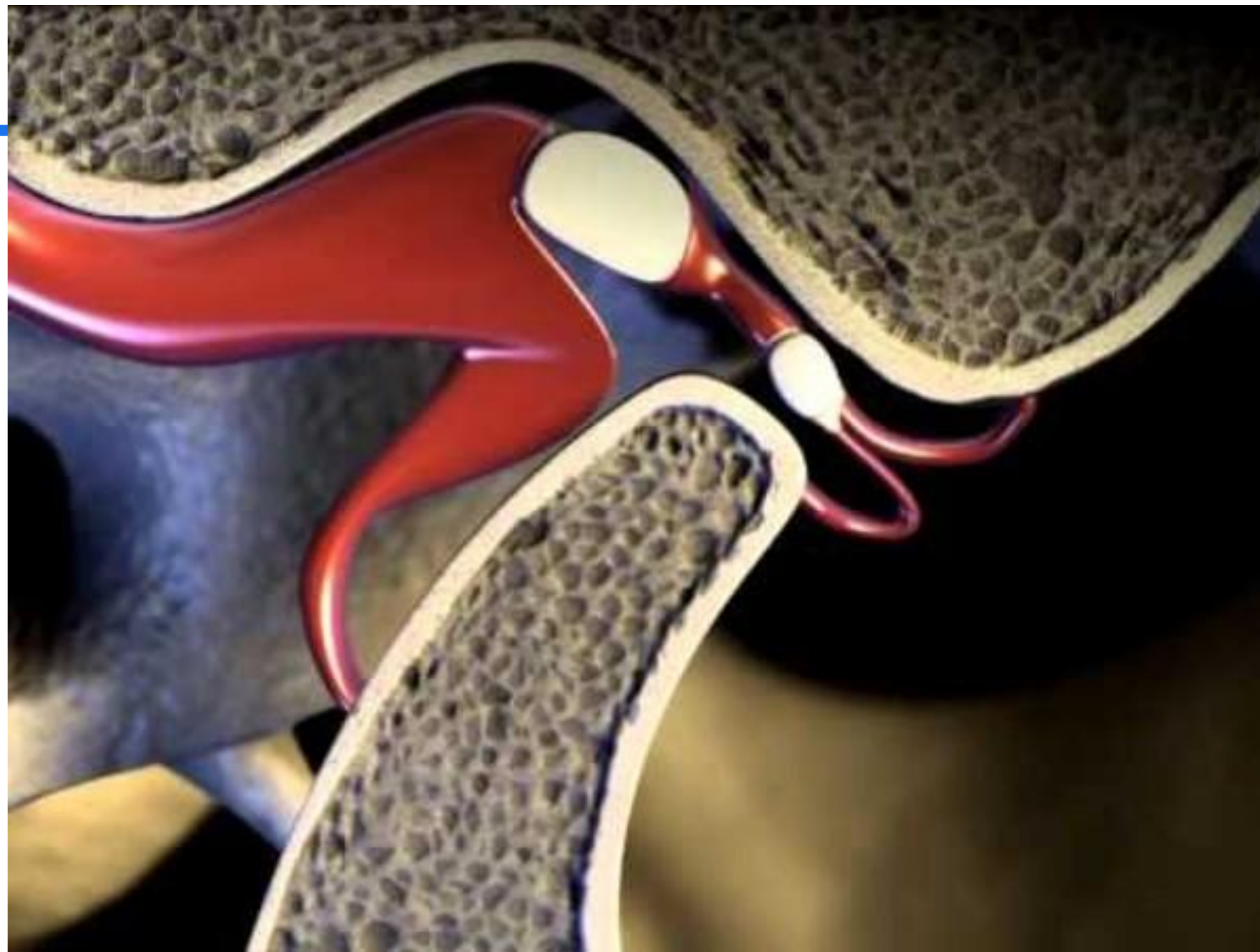
Joint Capsule of the

TMJ

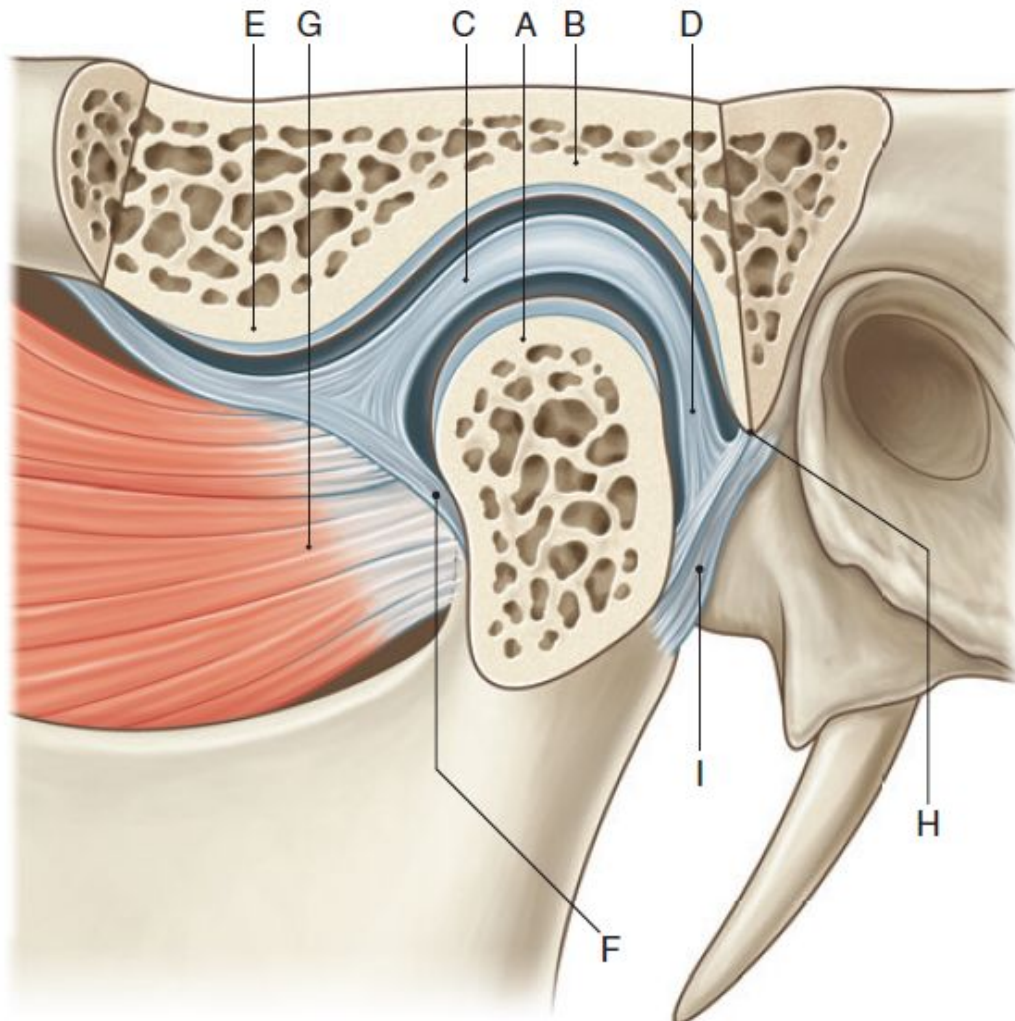
- **Histological Layers of the Capsule**
- **Outer Fibrous Layer**
 - Dense **fibrous connective tissue**
Reinforced by TMJ ligament
 - Provides **mechanical support and stability**
- **Inner Synovial Membrane**
 - Thin **vascular connective tissue**
 - Secretes **synovial fluid** → lubricates the joint and nourishes the **avascular disc**







Attachments of the TMJ Articular Disc

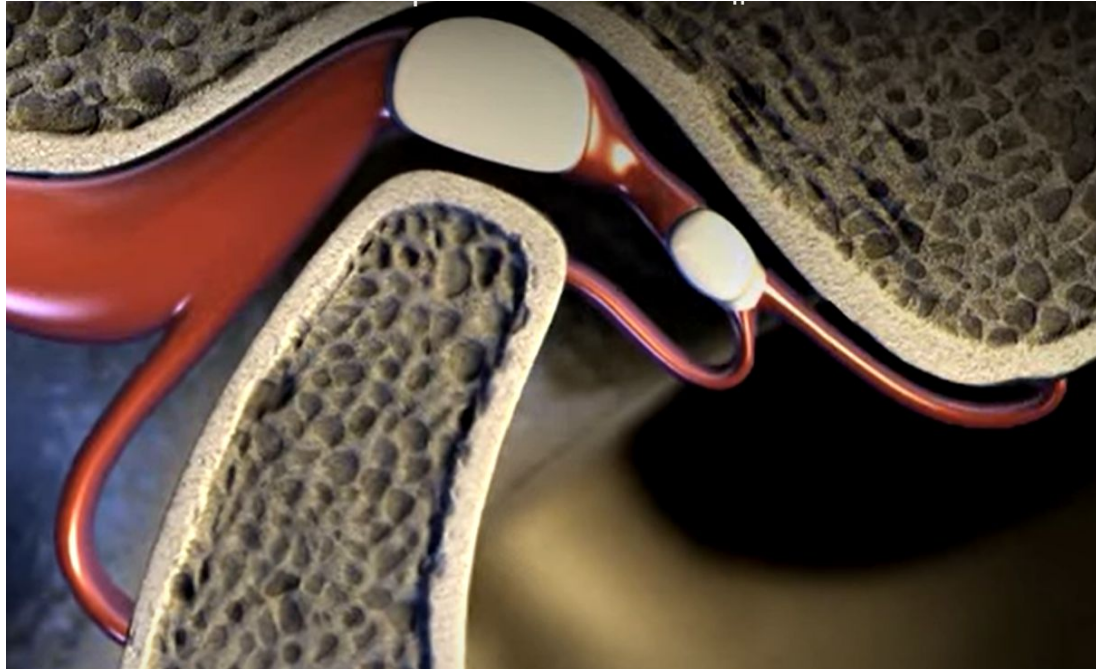


Lateral & Medial Attachments

- The disc is **firmly attached** to the **lateral and medial poles of the mandibular condyle to move with it.**

Anterior Attachment

- **indirectly via the joint capsule.**



Posterior Attachments

- Upper Posterior Lamina: Attached to the postglenoid process of the temporal bone.
- Lower Posterior Lamina: Attached to the neck of the mandibular condyle.

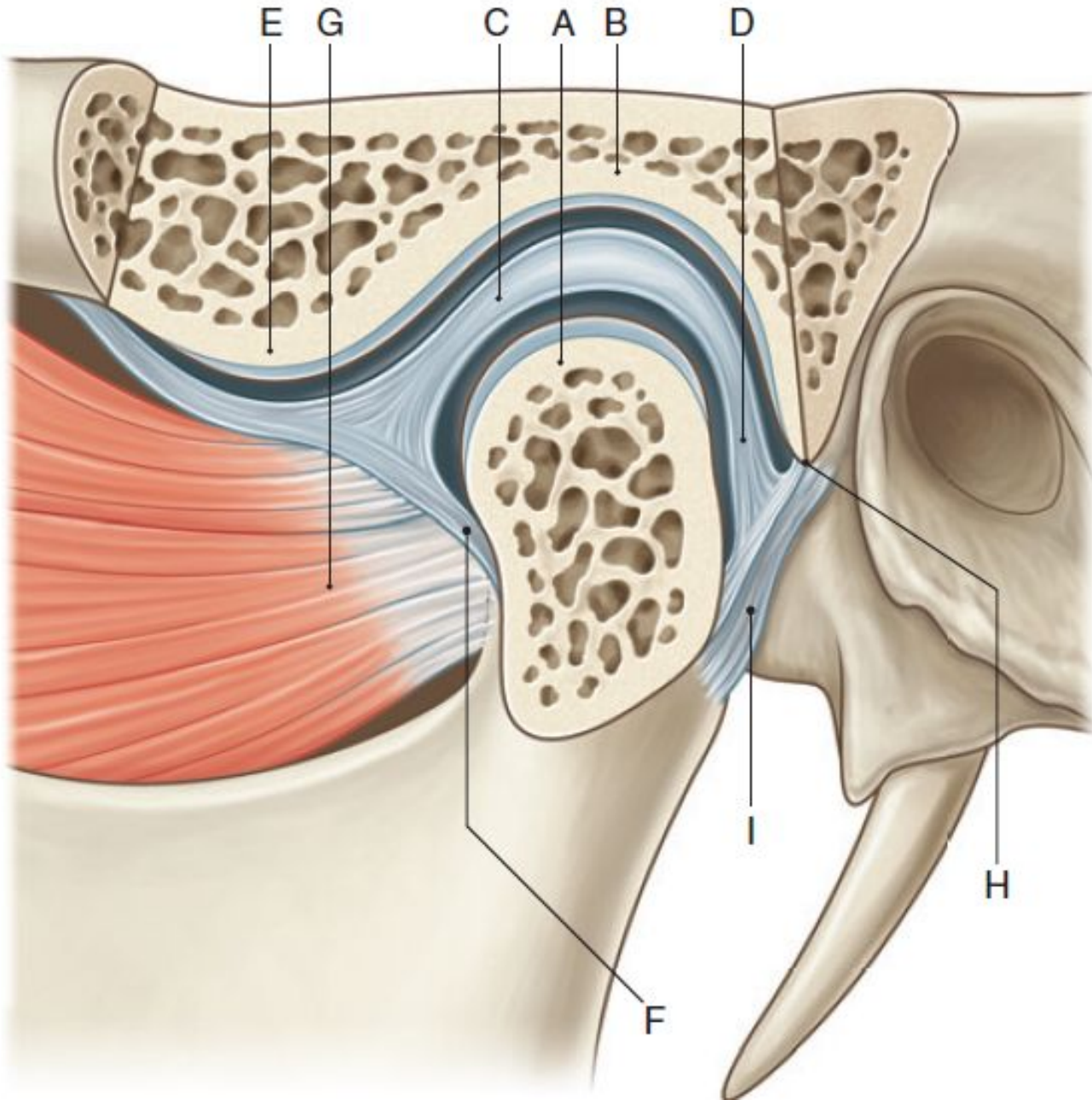


Fig. 3.8 Diagrammatic representation of the intra-articular disc.

A = mandibular condyle;

B = mandibular fossa;

C = central part of intra-articular disc;

D = posterior part of intra-articular disc;

E = articular eminence;

F = attachment of anterior part of intra-articular disc to anterior margin of mandibular condyle;

G = lateral pterygoid muscle;

H = attachment of upper part of bilaminar zone;

I = attachment of the lower part of bilaminar zone.

Attachments of the TMJ Articular Disc

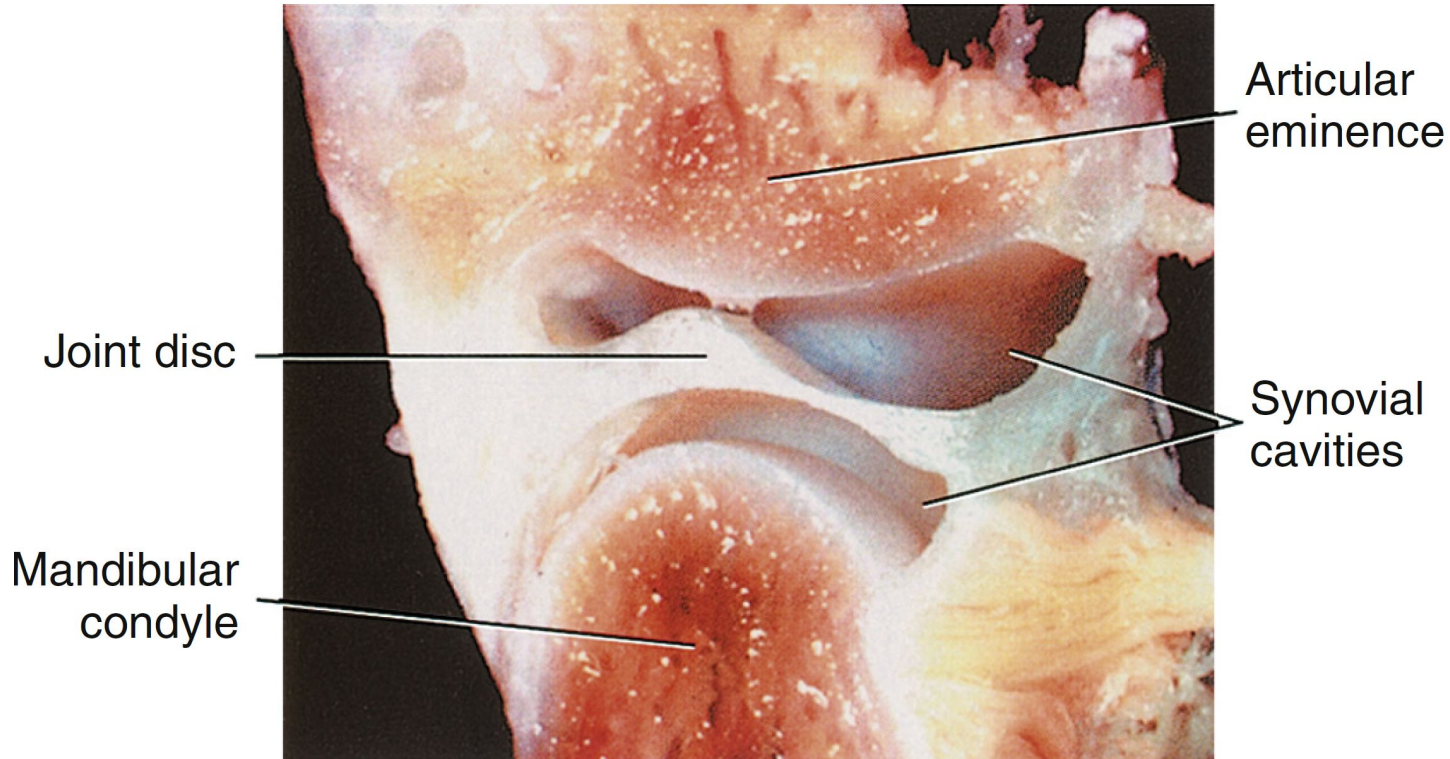


FIGURE 19-2 Block dissection of the temporomandibular joint. (From Nanci A: *Ten Cate's oral histology*, ed 8, St Louis, 2013, Mosby/Elsevier.)

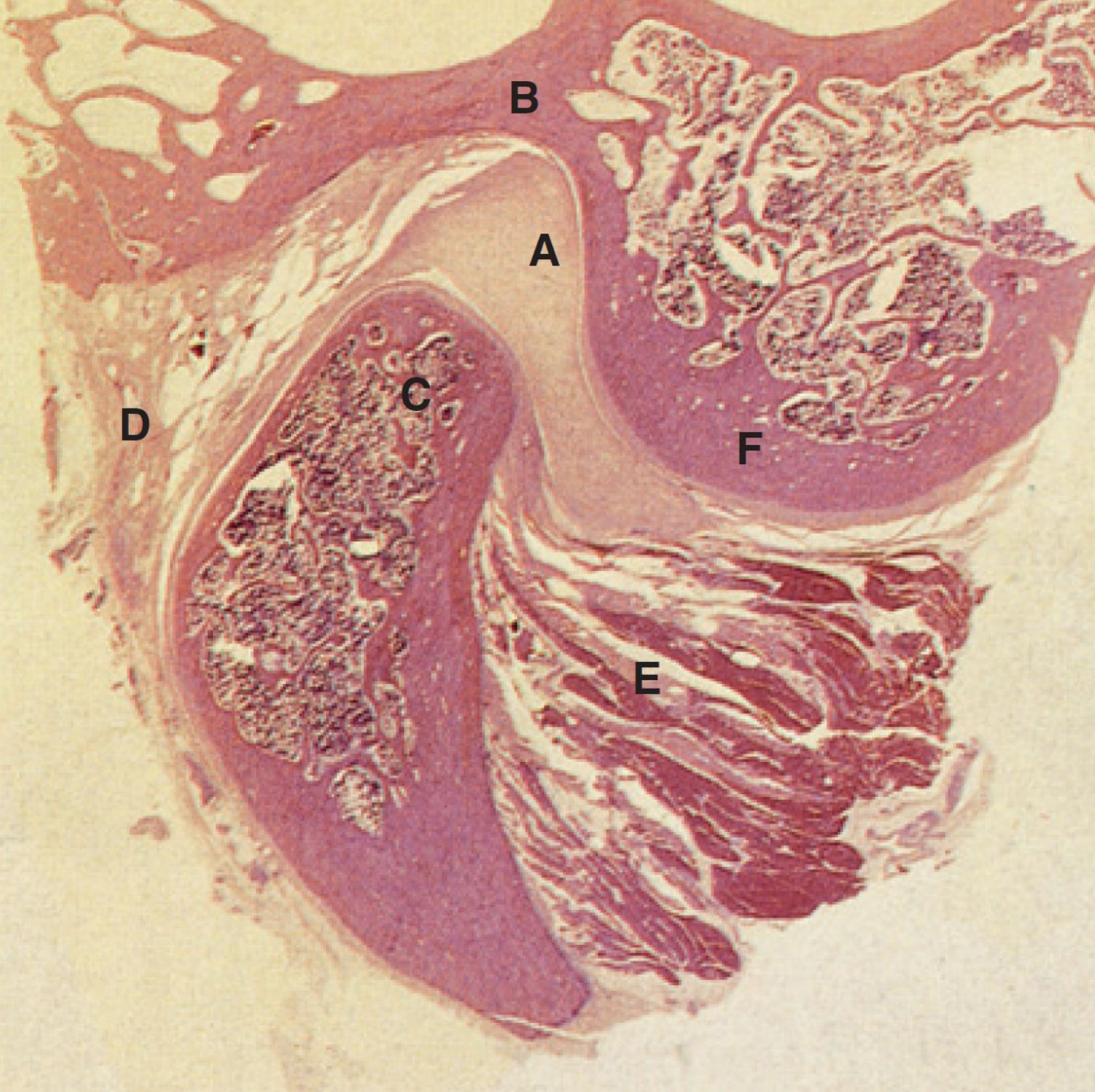


Fig. 15.1

Low-power sagittal section showing general distribution of tissues of the temporomandibular joint.

A = intra-articular disc;

B = mandibular (glenoid) fossa;

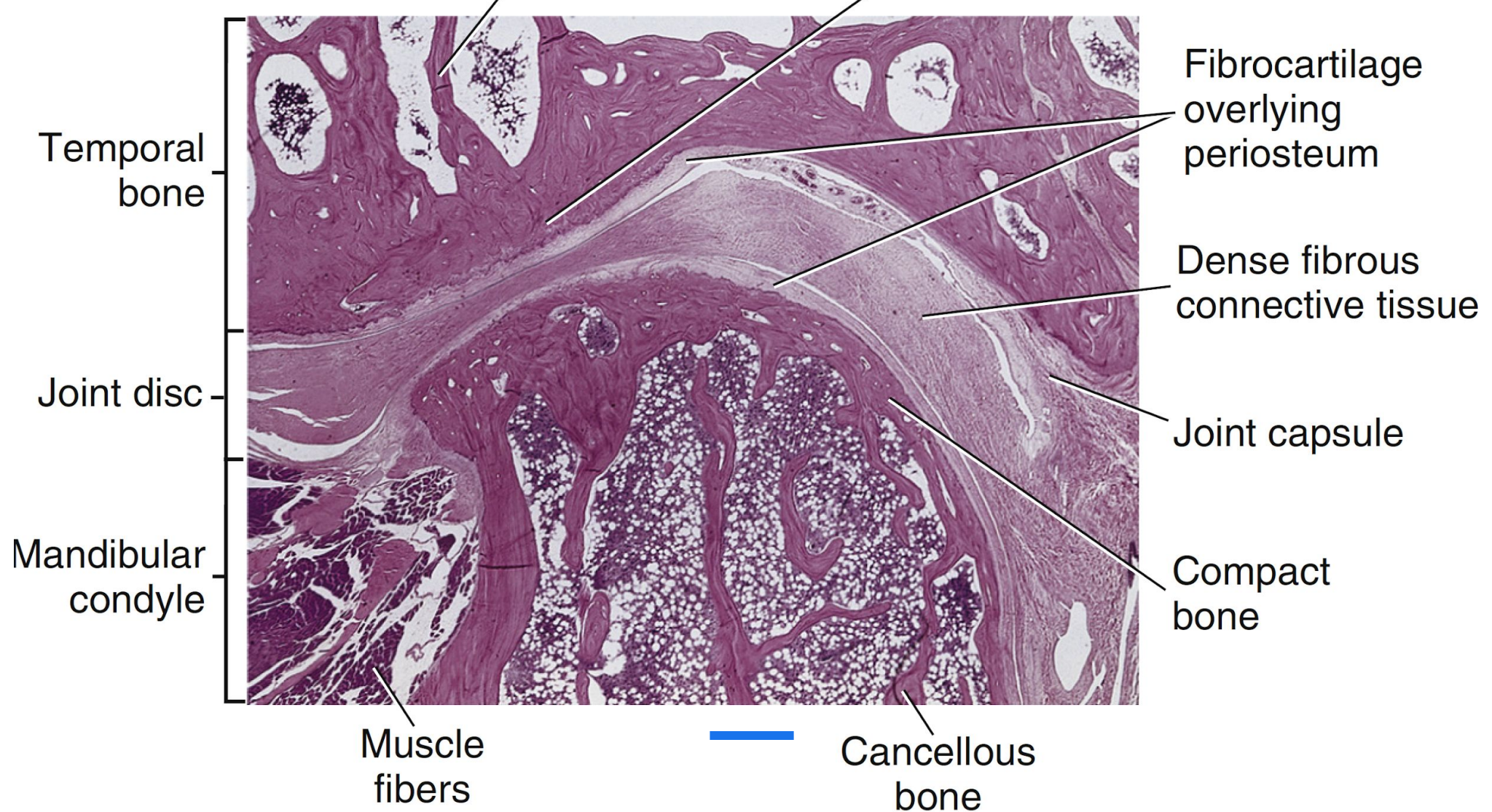
C = condyle of mandible;

D = capsule of joint;

E = lateral pterygoid muscle;

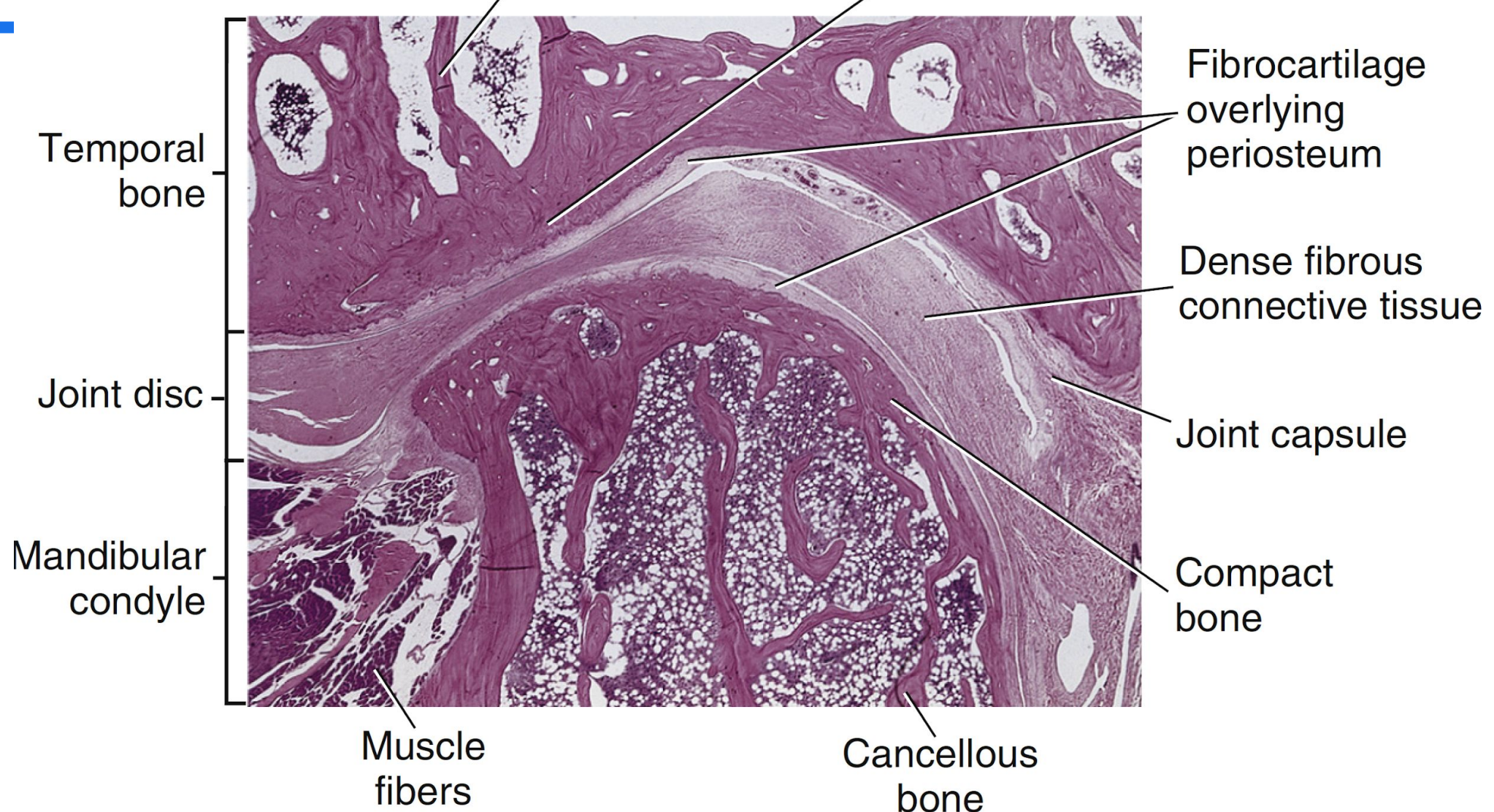
F = articular eminence

Identify mandibular condyle, mandibular fossa and articular eminence

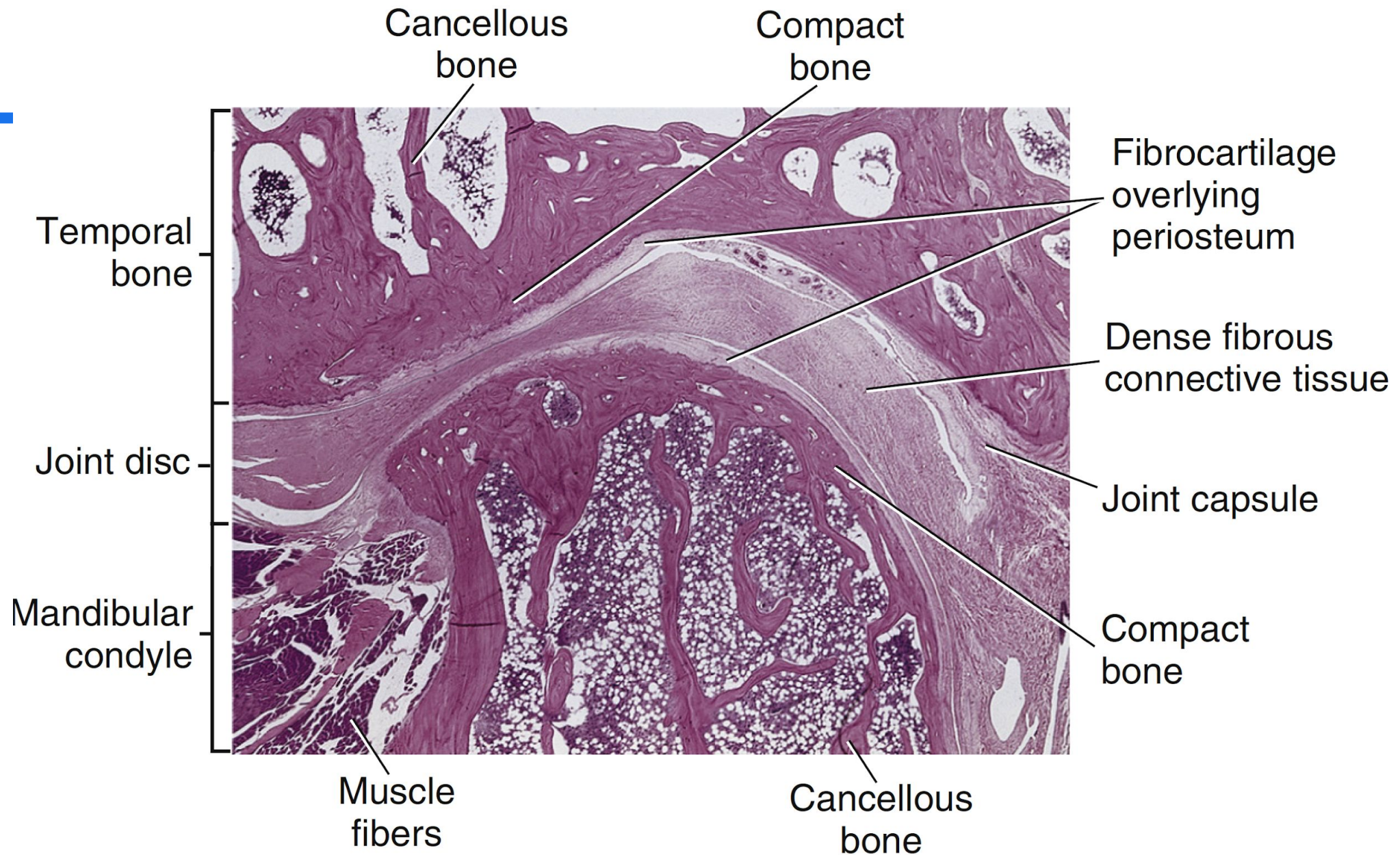


Articular Surfaces of the TMJ – Overview

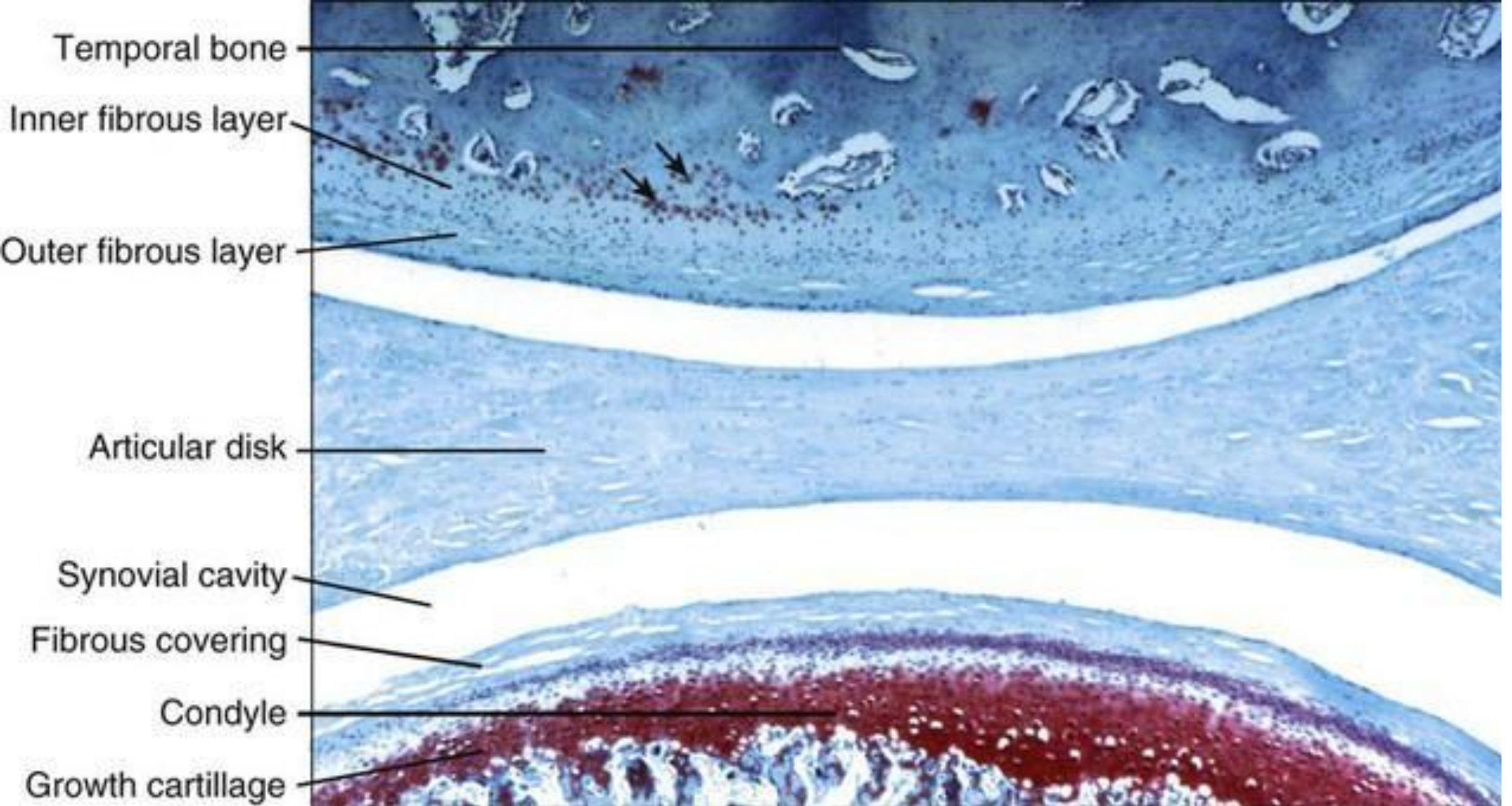
Identify lateral pterygoid Muscle attachment



Articular Surfaces of the TMJ – Overview



Articular Surfaces of the TMJ – Overview



Articular Surfaces of the TMJ – Overview

Articular Surface of the Mandibular Condyle (Adult)

Four distinct histological layers:

1. Superficial Fibrous articular surface Layer; B

Tissue: Dense fibrous connective tissue, **collagen type 1**, some **elastin**

Cells: Sparse **fibroblasts/fibrocytes**

Orientation: Collagen fibers **parallel** to surface (wavy/crimped under polarized light)

2. Cell-Rich Zone: C

Contains **proliferating cells** for renewal,

3. Fibrocartilaginous Layer: D

Tissue: Dense connective tissue with **rounded cartilage-like cells**, like **chondrocytes**

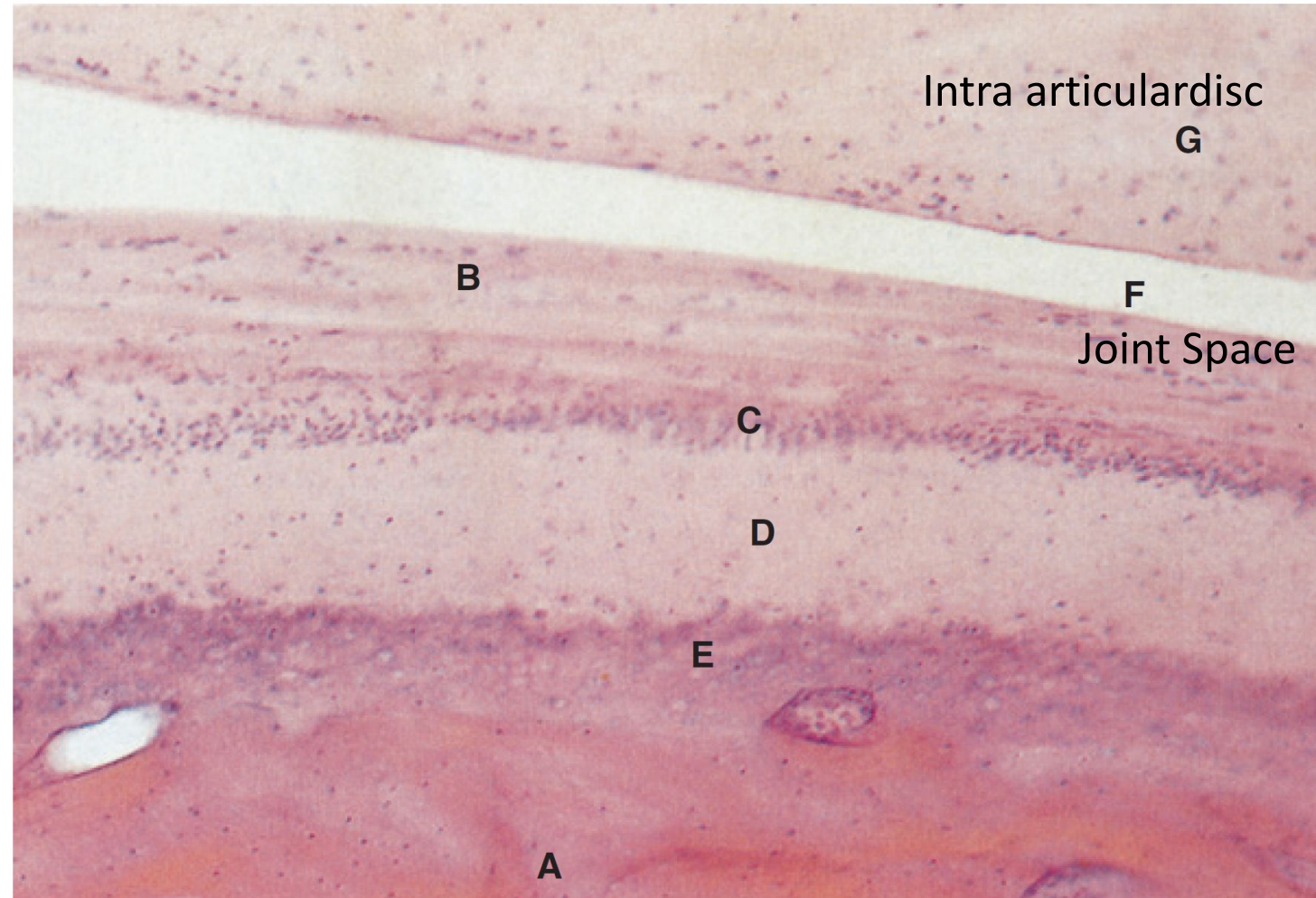
Appearance: resembles **fibrocartilage**

4. Zone of Calcified Cartilage: E

Location: Just above subchondral bone

Origin: Remnant of **secondary condylar cartilage**

Staining: Distinct from bone on histological staining



condyle

Articular Surface of the Mandibular Fossa



- The articular surface of the mandibular fossa is similar in composition to that of the mandibular condyle.
- It is generally thinner than the condylar surface.
- Becomes thicker as it passes over the articular eminence, where functional loading increases.
- Superficial collagen fibers exhibit crimping (wavy appearance), which provides elasticity and resistance to shear forces during joint movement.

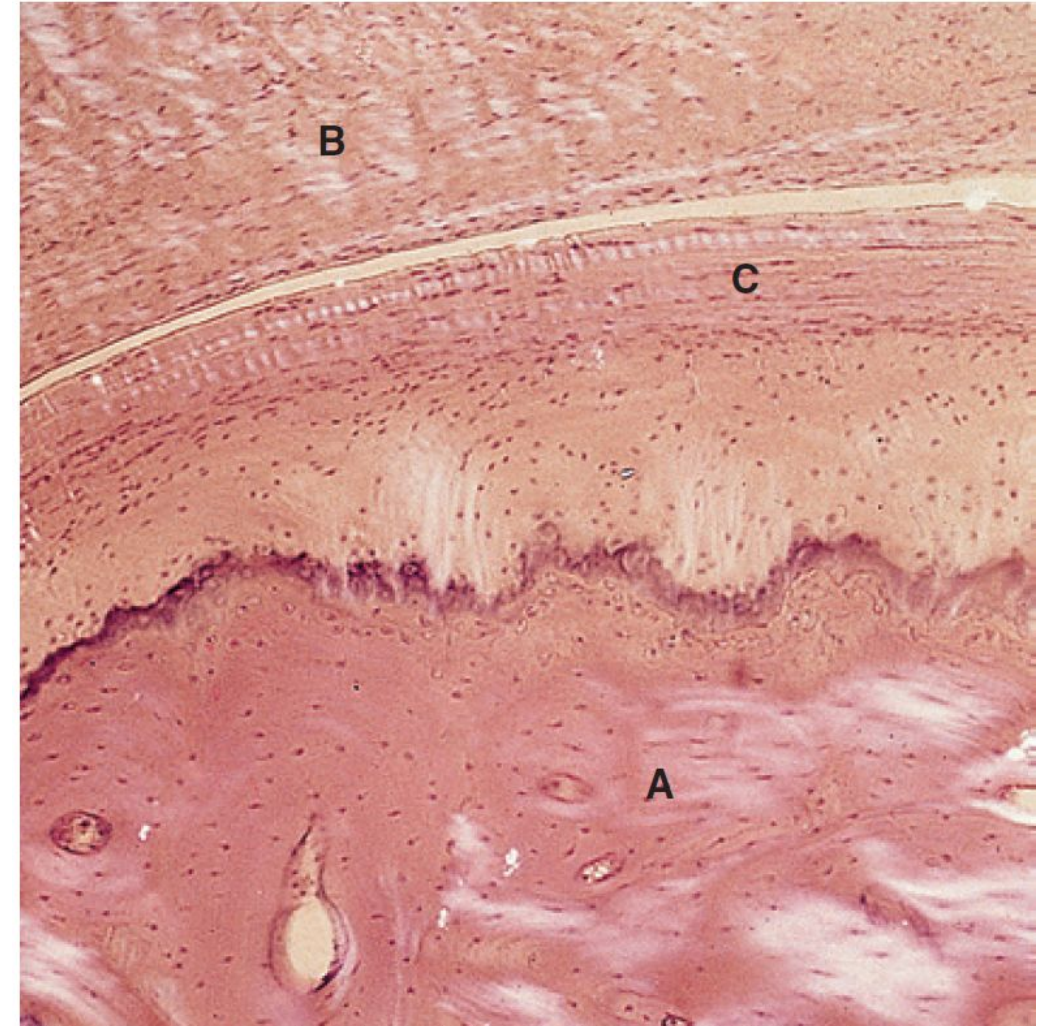
Articular Surface of the Mandibular Condyle (Adult)

- **Fig. 15.3** Sagittal section of head of adult condyle

A. Head of adult condyle

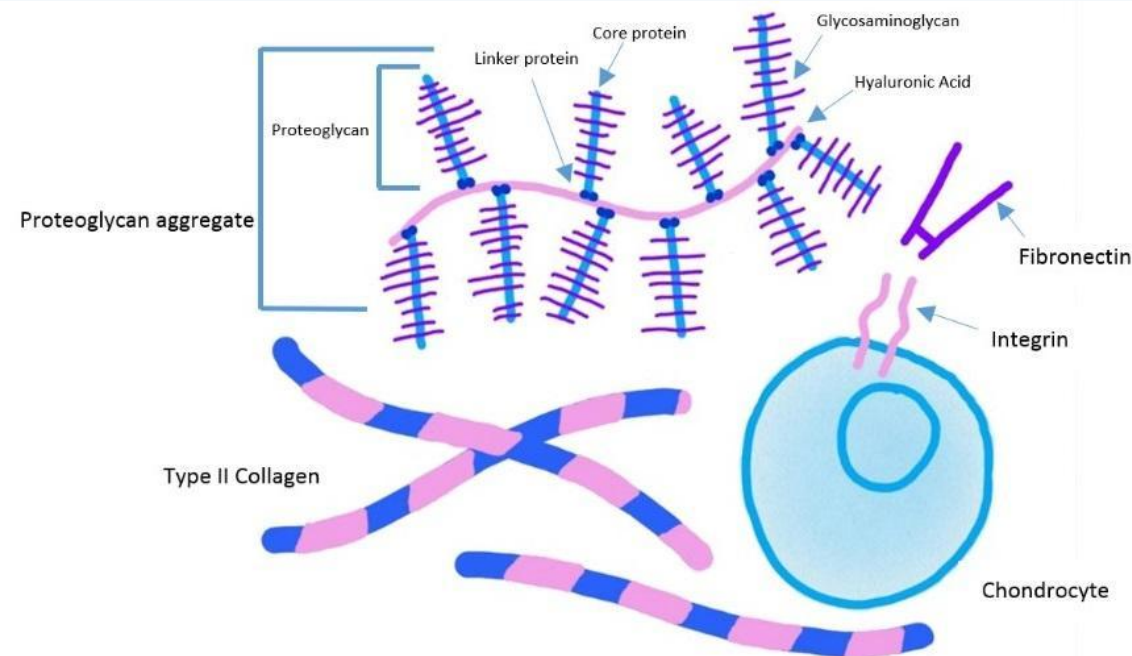
A. lower part of intra-articular disc

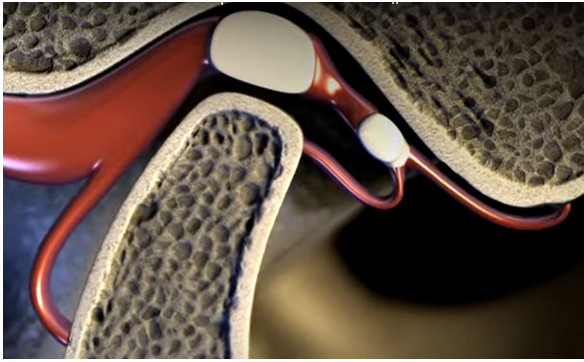
B. fibrous articular layer of the condyle



Box 15.1 Composition of intra-articular disc

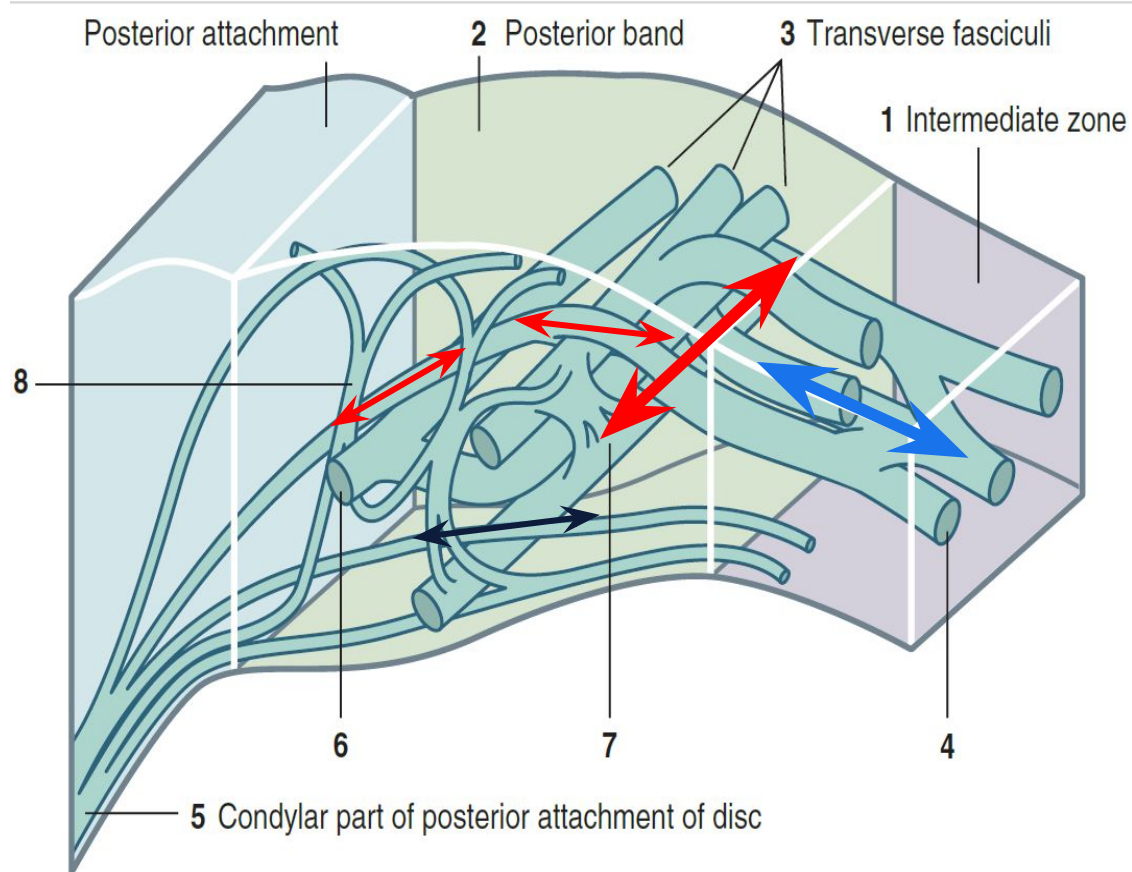
- 1) Water: 65–80%
- 2) Macromolecules: 20–35%
 - Collagen: 70–85%
 - Cells: 10%
 - Extracellular matrix components: 10% – e.g., proteoglycans, glycoproteins (fibronectin), glycosaminoglycans (chondroitin sulphate), elastin



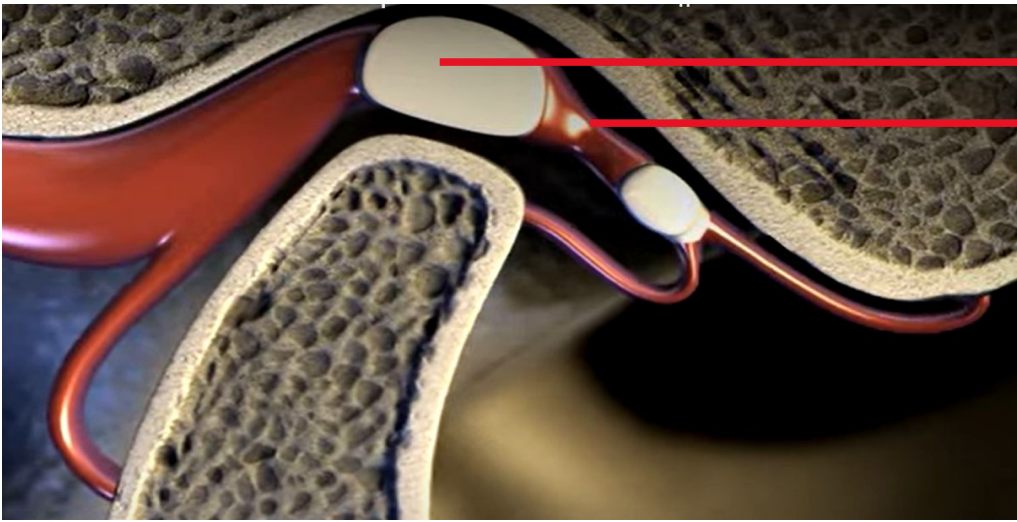


Collagen Fiber Arrangement in the TMJ

Disc is composed mainly of **type I collagen** (~80% of dry weight);



- **Traces** of types II, III, VI, IX, XII.
- **Intermediate zone:** fibers run anteroposteriorly.
- **Anterior/posterior bands:** fibers run transversely, vertically, and obliquely.
- **Peripheral fibers:** arranged circumferentially.
- **Fiber bundles** branch, decussate, and join.
- **Crimping seen in superficial fibers** → indicates compressional and tensional loading.



ELASTIN FIBRES

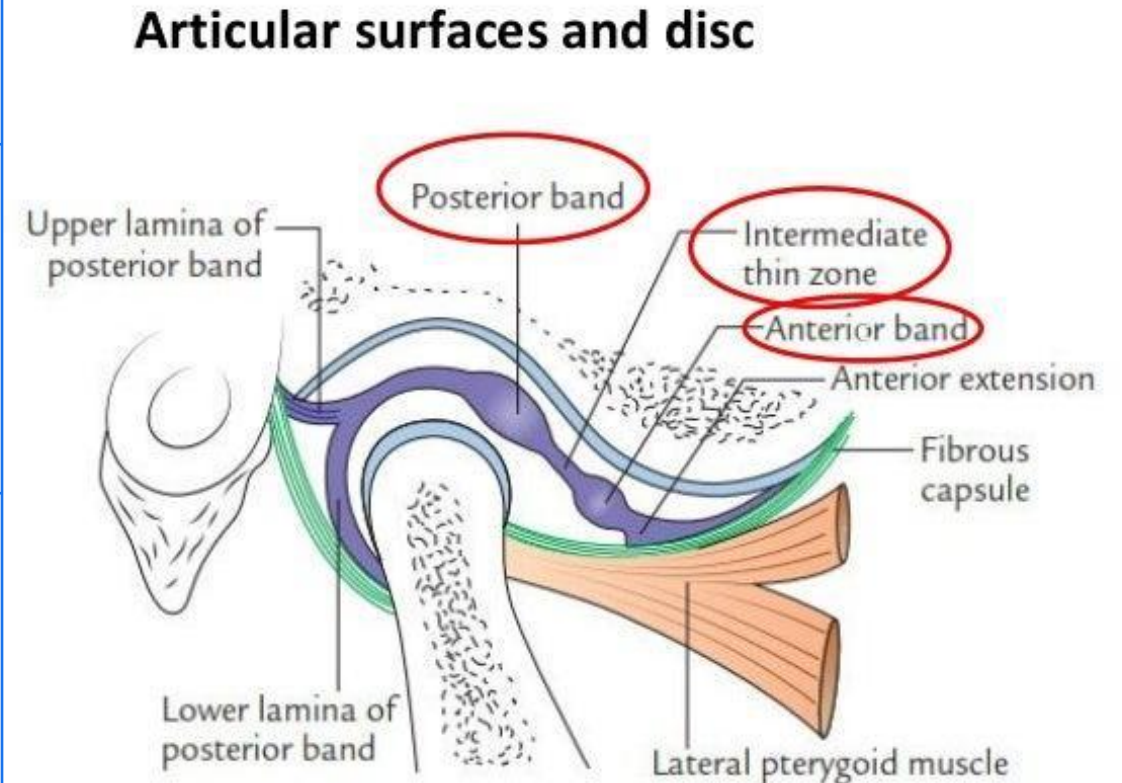
- ~5% of dry weight;
- decrease with age.
- Prominent in upper bilaminar zone, more lateral than medial.
- Present in anterior/posterior bands,
- absent in intermediate zone.
- Function: Disc recoil after loading / displacement.

GROUND SUBSTANCE

- ~5% of dry weight;
- richest in intermediate zone, more medially.
- Main GAGs: chondroitin sulfate, dermatan sulfate → absorb water, resist compression.
- Forms large proteoglycans (e.g., aggrecan-like).
- Fibronectin & tenascin also present.

TMJ Articular Disc (Posterior & Intermediate Zones) – Biochemical Link

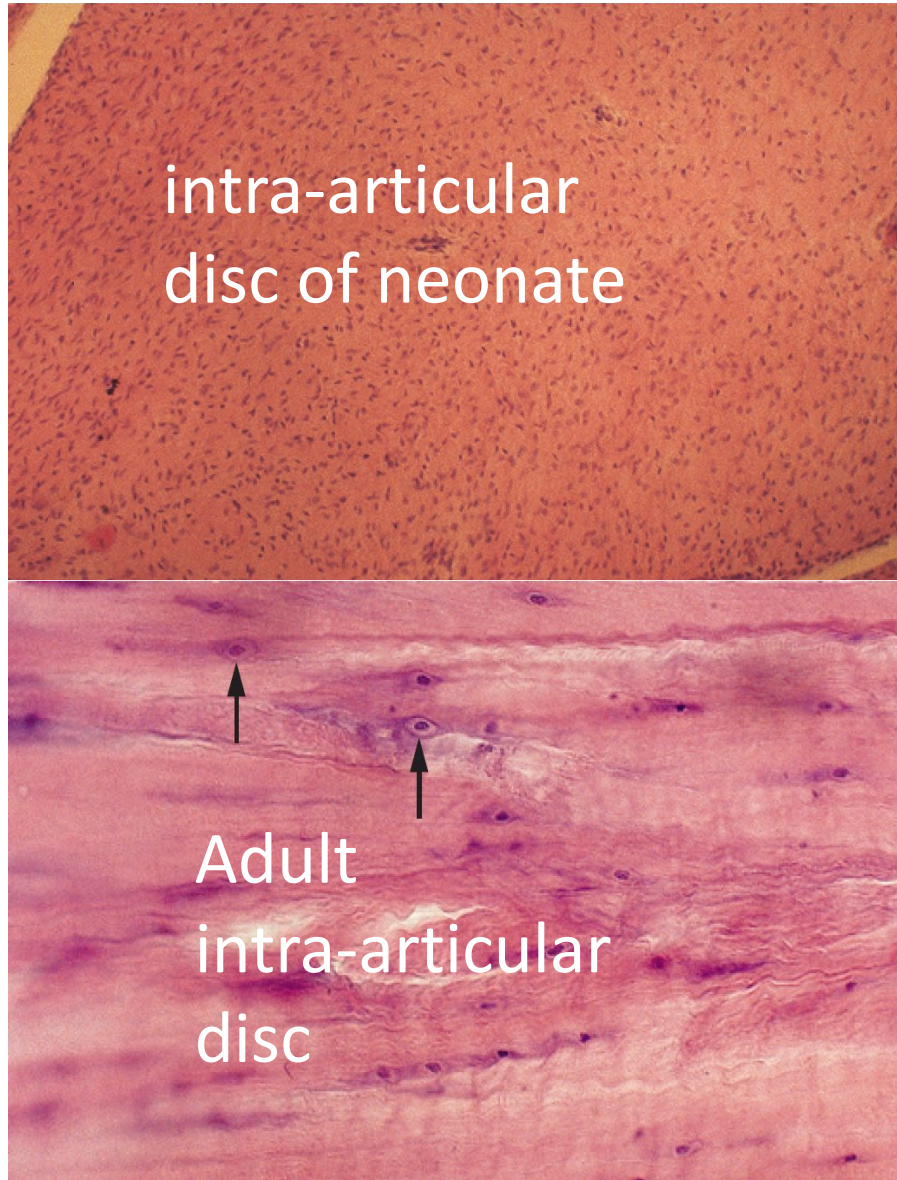
Zone	Main Molecules	Function
Intermediate zone	Collagen ↑, Proteoglycan ↓	Strength + load-bearing
Posterior zone	Proteoglycan ↑, GAG ↑, Glycoprotein ↑	Shock absorption + elasticity + repair



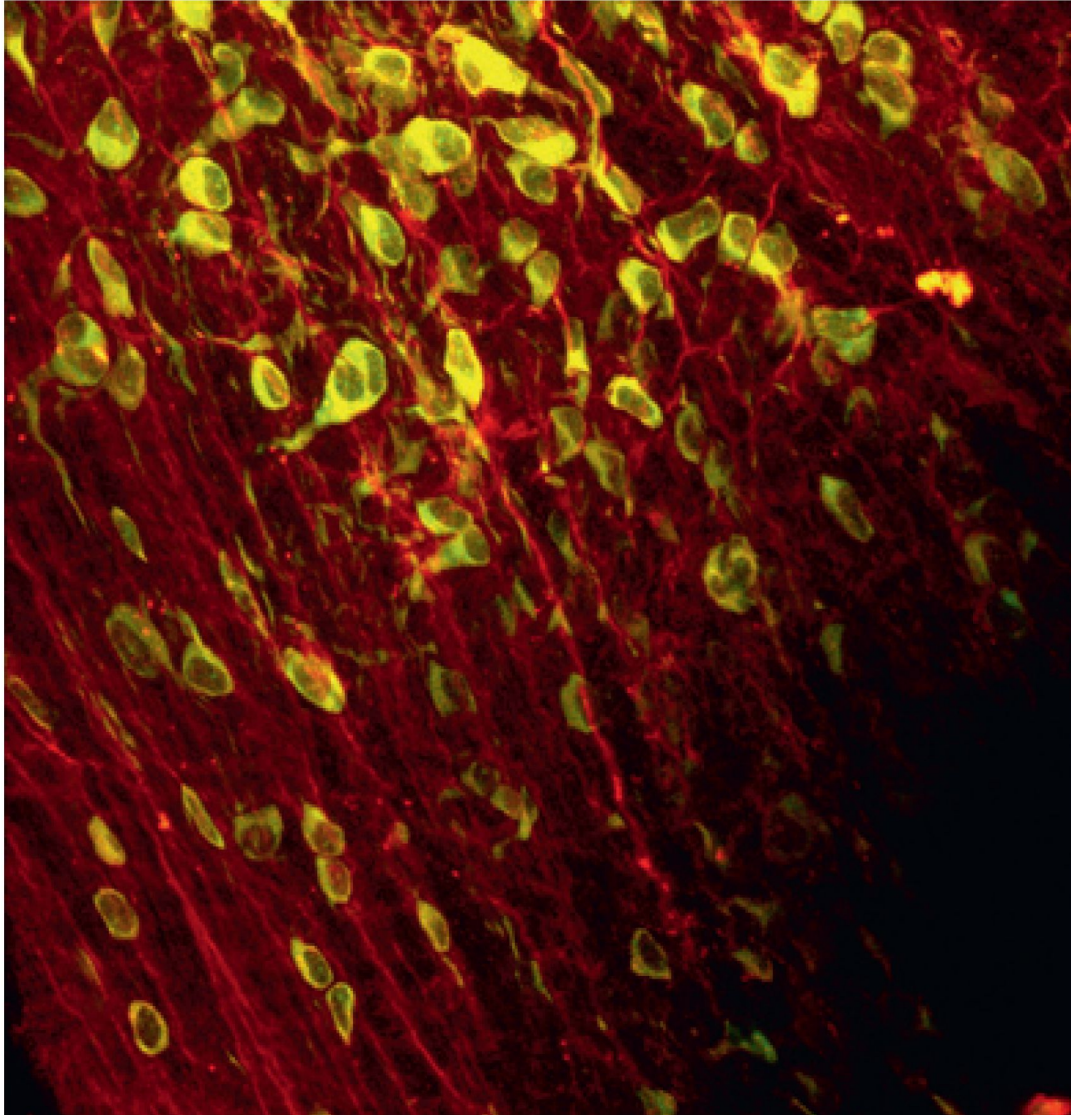
• **Intermediate Zone** = Strong (collagen)

• **Posterior Zone** = Soft & Vascular (proteoglycan + GAG + glycoprotein)

Cells of the TMJ Intra-articular Disc



- More **numerous at birth**, become **sparse in adults**.
- Cell types: Flattened (fibroblast-like) and rounded (chondrocyte-like).
- Contain moderate organelles → ECM synthesis; sometimes called fibroblasts/fibrocytes (ECM embedded fibroblast).
- Pericellular matrix mostly absent, seen only in older animals (no capsule).



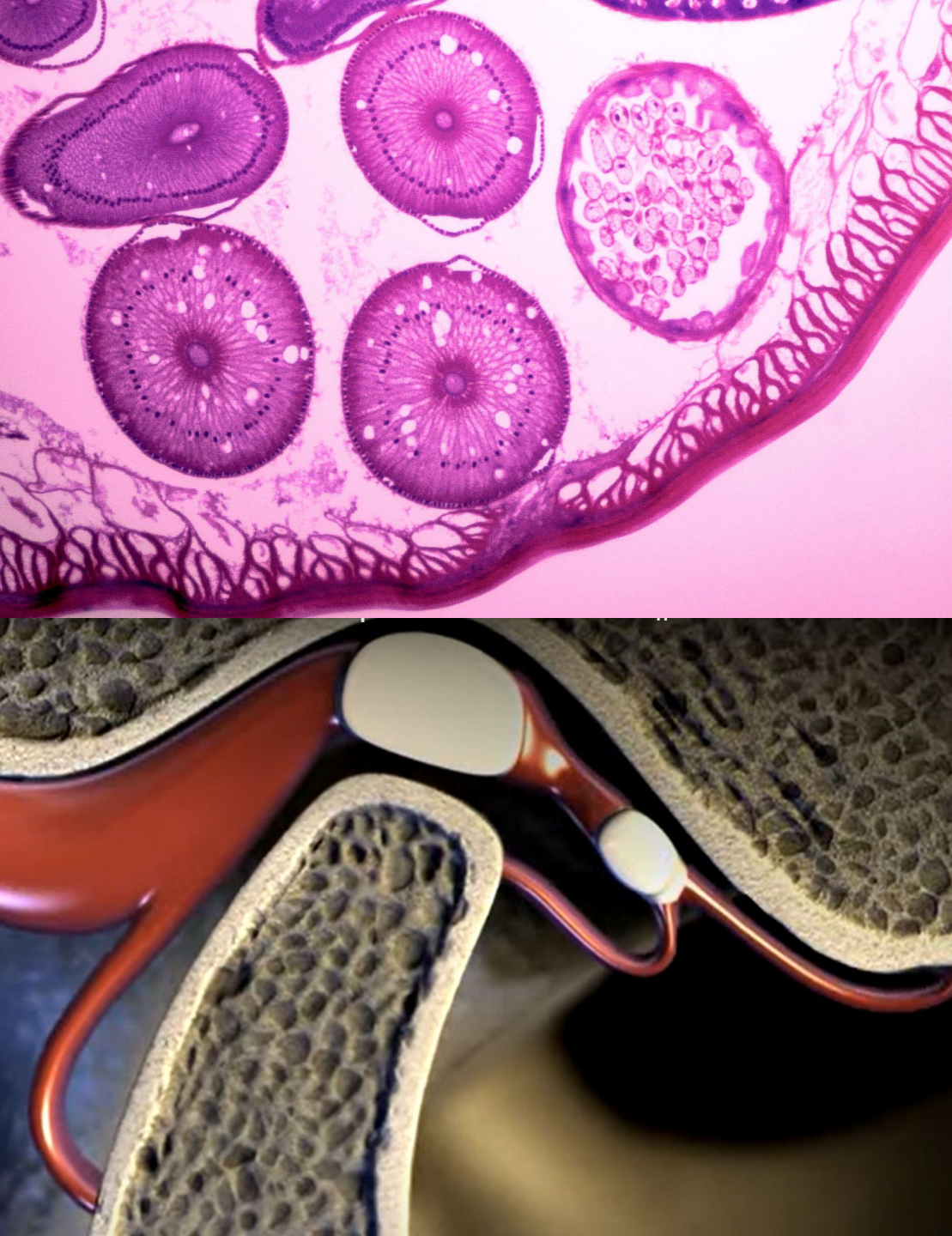
Confocal Image of TMJ Disc Cells (×250)

- **Green (Phalloidin):** Highlights cell bodies and long, fine cytoplasmic processes extend up to 100 μm . (F-actin).
- **Red (Connexin 43 antibody):** Labels gap junction protein within the cell processes.
- These processes likely facilitate diffusion of nutrients and fluids from peripheral blood vessels to the avascular centre (IZ) of the disc.

Box 15.2 Features associated with cartilage

- the presence of type II collagen **Only traces**
- the presence of high-molecular-weight chondroitin-sulphate-containing proteoglycans
- the presence of a pericellular matrix surrounding the cells
- chondrocyte-like cells that exhibit a rounded morphology with short, microvillus-like processes **Not true chondrocytes**
- the absence of cell contacts
- the absence of blood vessels within the tissue

Cells are often termed **fibrochondrocytes** due to **mixed fibroblast–chondrocyte features**.



Vascularization of TMJ Disc

- ❖ **Birth:** Vascular → **Adult:** Avascular, vessels at periphery.
- **Superior lamella (posterior):**
Vascular, venous engorgement during jaw opening (4–5× volume).
- ✓ Linked to: Pterygoid venous plexus.
- ✓ Jaw closure: Blood drains with elastic recoil
- **Inferior lamella (posterior):**
Avascular, inelastic.

Synovial membrane

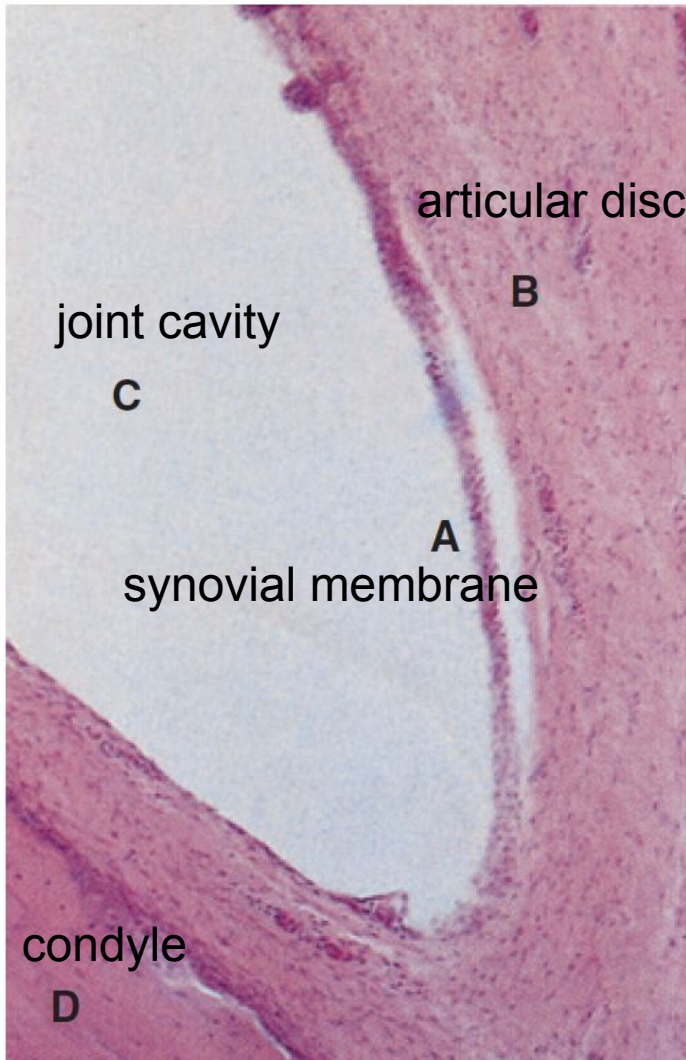


Fig. 15.20

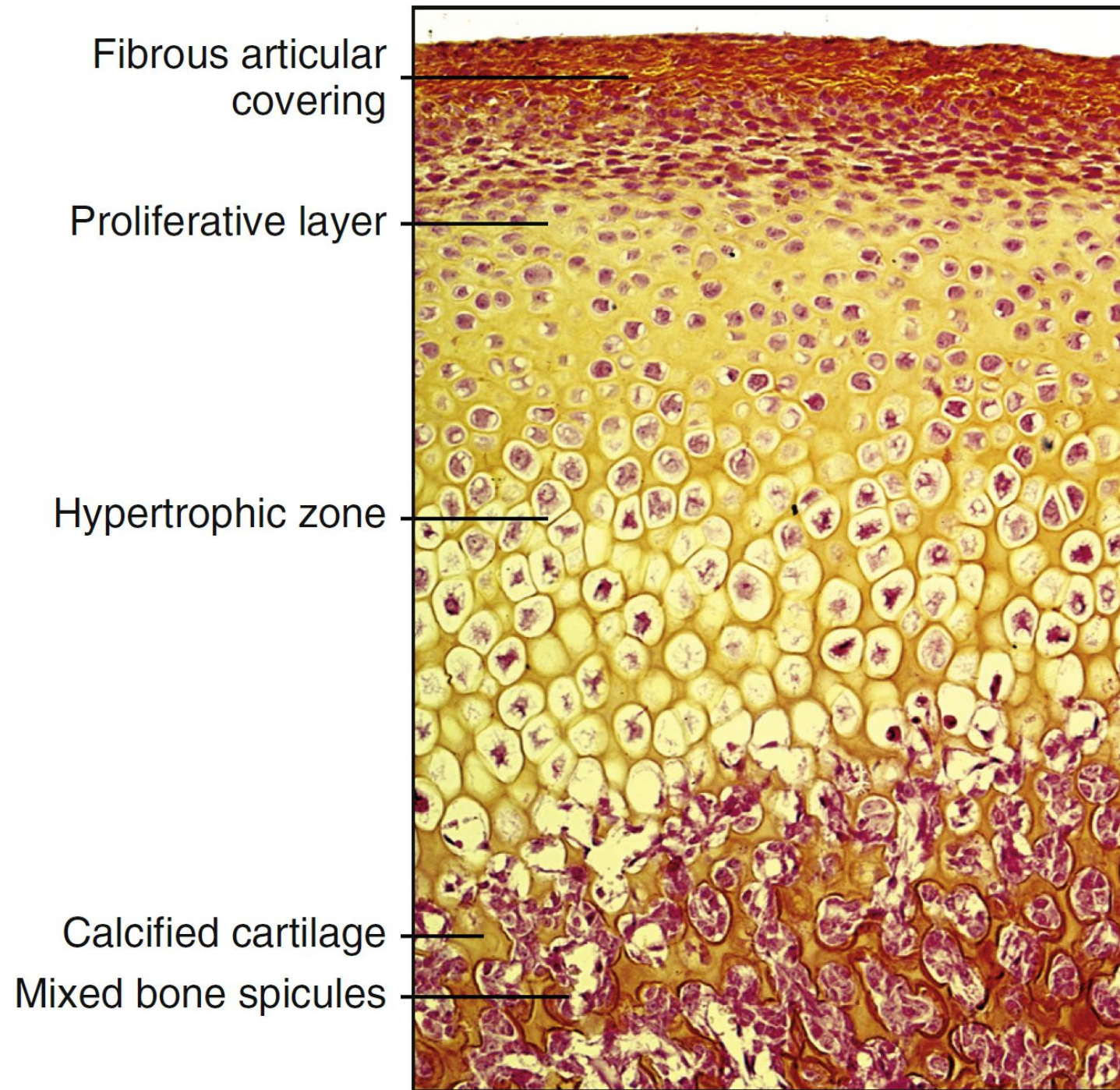
- **Lines:** Inner fibrous capsule + disc margins (not articular surfaces).
- **Active:** High metabolism, regenerative; **folded at rest**, flattens with movement.
- **With age:** Projections **increase** in size and number.
- **Structure:**
 - **Type A cells:** Macrophage-like
 - **Type B cells:** Fibroblast-like (secrete synovial fluid)
- **Synovial fluid:**
 - **Inferior cavity:** ~1 mL
 - **Superior cavity:** Slightly more

Synovial Fluid

- Secreted by:** Synovial membrane; fills joint cavities.
- Pressure:** Subatmospheric at rest, elevated during mastication and in bruxism.
- Functions:** Lubrication, Nutrition for avascular disc & articular surfaces
- Key Components:**
Water (major), Hyaluronan → main lubricant, **Lubricin (proteoglycan 4)**,
Phospholipids, Bioactive molecules (e.g., TGF- β , VEGF, MMPs, TIMPs, RANKL, ROS, IgA, IgG)

Articular Surface of the Mandibular Condyle (Child)

FIGURE 19-4
Growth center of mandibular condyle with endochondral transformation of hyaline cartilage into the alveolar process using interstitial growth. (From Nanci A: *Ten Cate's oral histology*.)

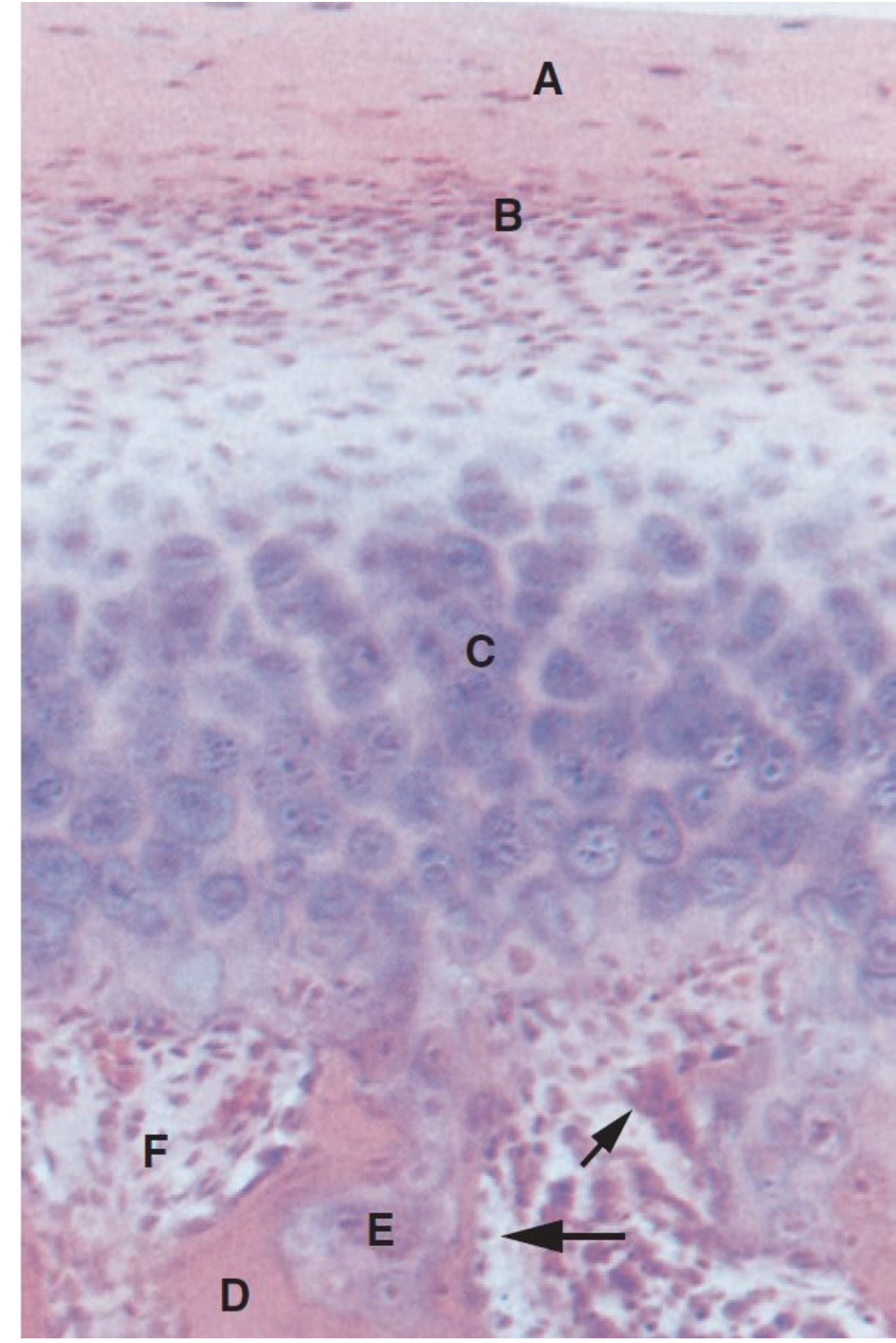


Articular Surface of the Mandibular Condyle (Child)

- **Four layers:**

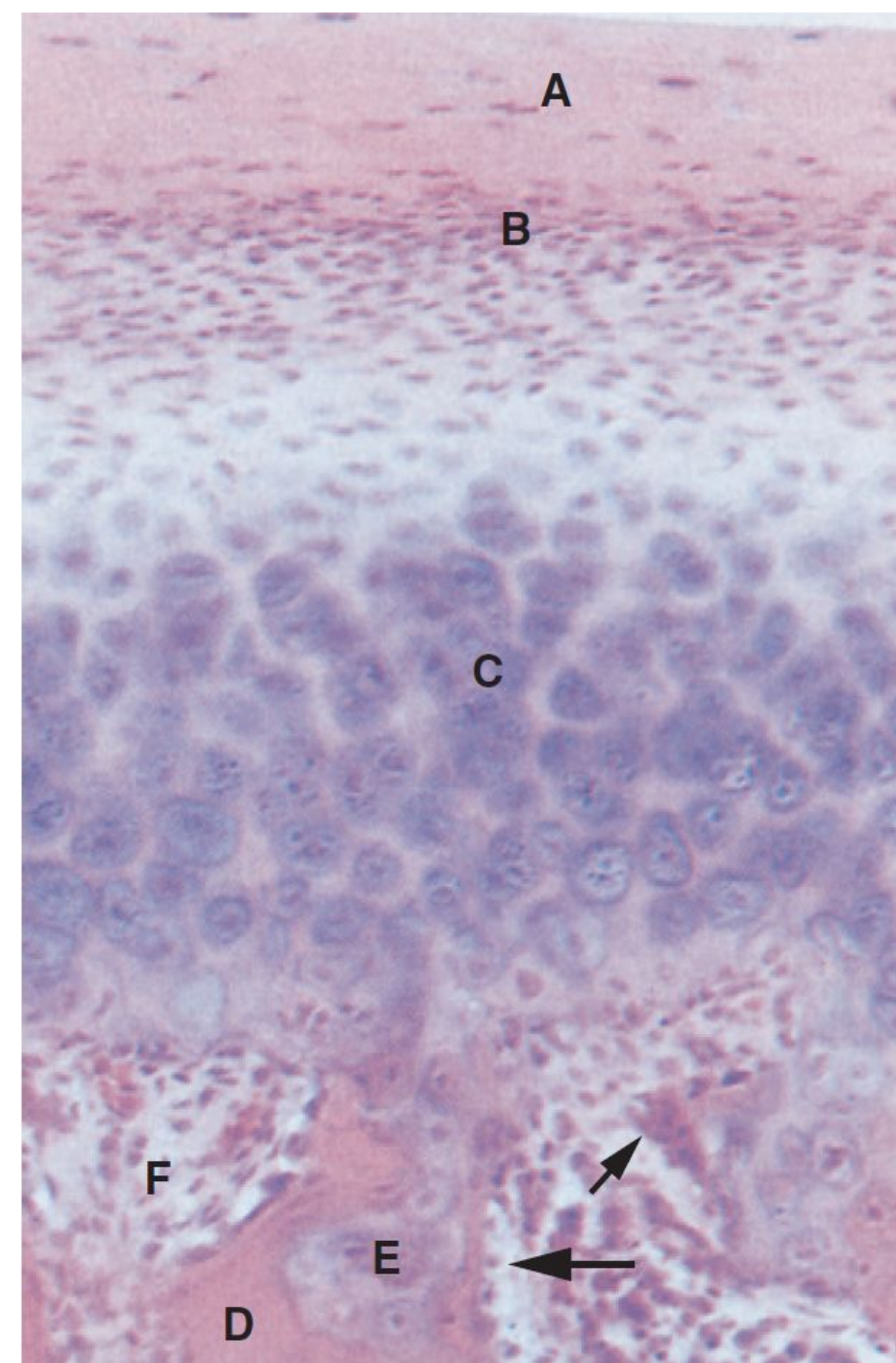
- A; Outer fibrous articular layer;
- B; cell-rich proliferative layer
- C; Hypertrophic chondrocytes of the secondary condylar cartilage
- D; woven bone being deposited around
- E; a template of calcified cartilage
- F; marrow space

✓ **Secondary condylar cartilage** appears ~10th week IU, active until adolescence.



Articular Surface of the Mandibular Condyle (Child)

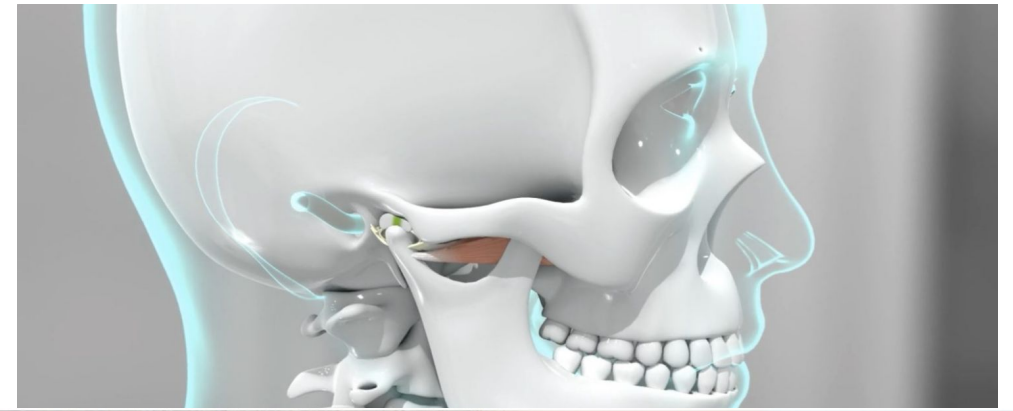
- **Type I collagen** in all layers; **Type II** in layers 3 & 4; **Type X** in hypertrophic zone.
- **Chondrocytes** undergo hypertrophy → **endochondral ossification**.
- **No columnar alignment** of chondrocytes (unlike growth plate).
- **Osteoblasts** form bone on **calcified cartilage template**; **osteoclasts** resorb cartilage.
- **No compact bone lining** (unlike adult condyle).
- **Mechanical force** influences differentiation



Feature	Child Condyle	Adult Condyle
Cartilage Type	Secondary condylar cartilage present	Replaced by bone; fibrocartilage on surface
Layers	4 layers: fibrous, proliferative, chondrocytic, hypertrophic	Mostly fibrous layer with reduced cellular activity
Growth Activity	Active endochondral ossification	No growth; ossified structure
Collagen Types	Types I, II, X	Primarily type I
Bone Lining	No distinct compact bone layer	Compact bone lining present
Cell Arrangement	Chondrocytes present but not in columns	No chondrogenic layers
Response to Force	Growth influenced by mechanical loading	Adaptive remodeling only

TMJ Disorders – Key Points

- **Multifactorial:** Common signs include **pain, clicking, limited mouth opening.**
- **Normal TMJ:** Low friction due to **synovial fluid** and **smooth surfaces.**
- **Disorders:**
 - **Increased friction** with degraded synovial fluid
 - **Disc displacement** (commonly anteromedial) → **abnormal loading**



The background of the slide is a dense, overlapping collage of numerous small, rectangular sticky notes. These notes are in various colors including teal, light blue, yellow, pink, and green. Each sticky note features a large, bold, black question mark. The notes are scattered across the entire frame, creating a textured and busy visual effect.

Thank you
Any questions
